Office of Research Assurance

Human Subject Research Protections Standard Operating Procedures (SOP)

SOP- 860: Reporting Adverse Events and Unanticipated Problems

Number	Effective Date	Version Number	Previous Versions
860	10/21/2024	v.1	N/A

Purpose

The information included within **ORA: SOP 860** outlines the requirements and responsibilities for reporting adverse events and unanticipated problems involving risks to subjects or others that may occur over the course of a research project. Events discussed in this document could be *expected* (e.g., bruising from a blood draw) or *unexpected* (e.g., theft of a laptop containing data) and may vary in *seriousness* and degree of *relation to the research*.

Principal Investigators (PI) and researchers must promptly notify the NASA IRB of adverse events and *unanticipated problems that involve risk to subjects or others* (i.e., UPIRSO) to ensure participant safety. Reports determined by the IRB to be UPIRSOs have additional reporting requirements (See SOP 810: Mandated Reporting).

> NOTE: For some HHS-conducted or -supported research, the Food and Drug Administration (FDA) and the HHS agency conducting or supporting the research (e.g., the National Institutes of Health [NIH]) may have separate regulatory and policy requirements regarding the reporting of unanticipated problems and adverse events. Anyone needing guidance on the reporting requirements of FDA or other HHS agencies should contact these agencies directly. Furthermore, investigators and IRBs should be cognizant of any applicable state and local laws and regulations related to unanticipated problems and adverse events experienced by research subjects, as well as foreign requirements for research conducted outside the United States. OHRP recommends that investigators and IRBs consult with their legal advisors for guidance regarding pertinent state, local, and international laws and regulations.

Types of events that require reporting to the NASA IRB include:

- Injury, disability, incapacity, hospitalization, life-threatening experience, death, side-effects, aggressive or unusual behavior, or other problem that was possibly related to research procedures, regardless of severity
- Risk of or Harm or damage to the safety, rights, or welfare of research participants, research staff, or others;
- Any accidental or unintentional change to the IRB-approved protocol that involved risks or has the potential to recur;
- Any publication in the literature, safety monitoring report, interim result, or other finding that indicates an unexpected change to the risks or potential benefits of the research;
- Any complaint of a participant that indicates an unanticipated risk or which cannot be resolved by the research staff;

- Breach of privacy, confidentiality, or data security;
- Loss or destruction of research data not in accordance with IRB-approved procedures;
- Possible abuse disclosed in an interview and/or survey;
- Urgent situations that may occur as the result of unanticipated problems during the research project

Scope

This policy applies to all non-exempt human subjects research under the oversight of the NASA IRB. Collaborative research involving Reliance Acknowledgements (RA) in which NASA cedes oversight to another institution follows the terms established in the RA. Projects receiving NASA funding, but under another IRB's oversight, are subject to the policy of the reviewing IRB and the terms established in the funding award.

Definitions

Adverse event: [1] Any untoward or unfavorable [medical] occurrence in a research participant [including any abnormal sign, symptom, or disease] that is temporally associated with the participant's involvement in the research. It may be anticipated (i.e., expected) or unexpected. An adverse event encompasses physical, psychological, social, economic, legal, or informational harms.

Events: In this SOP, events refer to both unanticipated problems and adverse events. An internal event is one that occurs with research participants enrolled in a project under the purview of or approved by the NASA IRB and/or directed by a NASA researcher. An external event is one that occurs with research participants enrolled in a multi-site project that do not fall under the purview of the NASA IRB.

Serious adverse event: [2] Any adverse event temporally associated with the individual's participation in research that meets *any* of the following criteria:

- Results in death.
- Is life threatening (places the subject at immediate risk of death from the event as it occurs).
- Requires inpatient hospitalization or prolongation of existing hospitalization.
- Results in a persistent or significant disability/incapacity.
- Results in a congenital anomaly/birth defect. or
- Any other adverse event that, based upon appropriate medical judgment, may jeopardize the subject's health that may require medical or surgical intervention to prevent one of the other outcomes listed in this definition.

Unanticipated problems involving risks to subjects or others (referenced hereafter as **Unanticipated Problems**): [3] Any incident, experience, or outcome that meets ***all*** the following criteria:

- Is *unexpected* (in terms of nature, severity, or frequency) given (a) the research procedures that are described in the protocol-related documents, such as the IRB-approved research protocol and informed consent document; and (b) the characteristics of the subject population being studied;

[1] "adverse event" is intended to encompass FDA's varied terminology including *adverse effect* [21 CFR 312.64] and *adverse experience* [§ 312.32] and *unanticipated adverse device effect* [21 CFR 812.3]
[2] (Office for Human Research Protections (OHRP), 2007)
[3] (Office for Human Research Protections (OHRP), 2007)

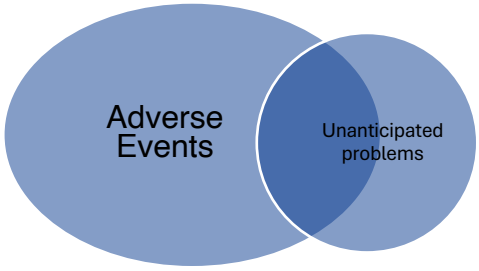
- Is *relate*d or *possibly related* to an individual's participation in the research (possibly meaning there is reasonable possibility the problem may have been caused by the research procedures); **and**
- *Suggests* that the research places subjects or others at *a greater risk of harm* (including physical, psychological, social, economic, legal, or informational harm) *than was previously known or recognized*.

For studies of medical devices, the FDA Investigational Device Exemption (IDE) regulations define:

Unanticipated adverse device effect (UADE): any serious adverse effect on health or safety or any life-threatening problem or death caused by, or associated with, a device, if that effect, problem, or death was not previously identified in nature, severity, or degree of incidence in the investigational plan or application (including a supplementary plan or application), or any other unanticipated serious problem associated with a device that relates to the rights, safety, or welfare of subjects (21 CFR 812.3(s)).

Not all Adverse Events are Unanticipated Problems

Not all Adverse events are Unanticipated Problems; similarly, not all Unanticipated Problems are Adverse Events.



Simply put, an adverse event is any harm that has occurred to a research participant that may or may not be directly related to the individual's participation in the research. Unanticipated Problems are those that have been determined to be related (or possibly related) to participation in the research.

While Unanticipated Problems are necessarily unexpected, adverse events include anticipatable occurrences in addition to those that are not. Designation as an Unanticipated Problem means that it suggests that the research presents greater risk of harm to subjects or others than previously known, regardless of whether harm has occurred.

Policy

(1) Researchers must conduct human subjects research in accordance with the protocol and materials as approved by the NASA Institutional Review Board (IRB), all applicable NASA policies, and federal regulatory requirements governing such research.

(2) It is the researcher's responsibility to promptly report all adverse events and unanticipated problems to the NASA IRB following established procedures[4] as outlined in this document. Instances involving actual or imminent harm to research participants or others must be reported to the IRB *immediately.* NOTE: Adverse Events/Unanticipated problems of drugs/devices under FDA oversight may have specified reporting deadlines.

(3) The NASA IRB reviews the report and makes determinations about whether the problem/event raises new concerns about:
   a. Risk to participants or others
   b. The risk/benefit ratio
   c. The approved informed consent form, and

[4] 14 CFR 1230.108(a)(4) and 21 CFR 56.108(b)

d. The need for re-consent

(4) The NASA IRB may also review the event for noncompliance according to SOP-850.

(5) External reporting. The Office of Research Assurance (ORA) ensures prompt reporting[5] to the Office of Human Research Protections (OHRP) within the Department of Health and Human Services (HHS) and, when applicable, the FDA and/or other federal agencies of IRB determined instances of:
   a. Unanticipated Problems Involving Risks to Subjects or Others (UPIRSO)
   b. Serious or continuing noncompliance [ORA: SOP-850]
   c. Any suspension or termination of IRB approval [ORA: SOP-820]

Procedures

Reporting Adverse Events or Unanticipated Problems to the IRB

1. Members of the research team must immediately report the event to the PI.
2. PIs are required, as a term of IRB approval, to *promptly* report Adverse Events or Unanticipated Problems to the NASA IRB.
   a. For clinical investigations of drug and biological products conducted under an Investigational New Drug (IND) application, PI are required to *promptly* report any adverse effect that may reasonably be regarded as caused by, or probably caused by, the drug.
      i. If the adverse effect is alarming, the investigator shall report the adverse effect immediately [§ 312.64(b)].
   b. For medical device studies, PI are required to submit a report of Unanticipated Adverse Device Effect (UADE) to the IRB and Sponsor as soon as possible, but no later than 10-business days after the PI first learns of the event [§ 812.150(a)(1)]
3. **Reporting Method.** Adverse events or Unanticipated Problems should be submitted as a **Report of New Information (RNI)** in the eIRB.
   a. PI or designated proxy may submit an RNI within the eIRB.
      i. An RNI may be initiated directly from the study workspace (subsequently linked to the main study) or as a generic RNI from the dashboard (may be later linked to relevant study records).
   b. PI or concerned individuals may also reach out directly to the IRB Chair, ORA Compliance Officer, or IRB staff IRB directly for guidance.
4. **Suggested content when reporting.** Sufficient information is needed to facilitate assessment of the issue and determination of the appropriate course of action.
   a. When possible, the report should include the following:
      i. Study Identification: PI name, study title, ID number,
      ii. Description of the problem, situation, or incident, including specific changes to procedures in the approved research plan if applicable
      iii. PI's assessment of why the event occurred.

[5] 14 CFR 1230.108(a)(4) and 21 CFR 56.108(b)

iv. Assessment of potential impacts (e.g., increased participant risk, reduced research benefits, or compromised research integrity)
v. Description of immediate changes to the protocol that were made in response to the situation, if any.
vi. Description of corrective actions that were implemented or that are proposed to prevent future similar occurrences.
vii. Relevant materials or documents

b. Ideally, the information in the report would allow the IRB to address whether the incident/event/problem was:
   i. expected (i.e., identified in the application and consent form) or unexpected.
   ii. related or possibly related to participation in the research.
   iii. suggest participants or others may be at greater risk of harm than was previously known or recognized.

Receipt of a report of potential noncompliance, complaint, or problems.

1. **Initial Assessment.** The ORA Compliance Officer or NASA IRB staff screen the report to determine whether it is complete. All complaints, problems, adverse events, deviations/noncompliance, and any new information suggesting changes to study risks/benefits are assessed by the IRB Chair or designee to ensure continued protection of research participants from avoidable harms.
2. **Chair Assessment.** The NASA IRB chair or designee will review the report to determine:
   a. If the event constitutes an ongoing risk of harm to currently enrolled or future participants or others
   b. How identifies risks of harm may be minimized
   c. Whether the project should be suspended or terminated (See SOP-820)
   d. Whether corrective actions or protection measures are required, and/or
   e. The presence of noncompliance (See SOP-850)
3. **Further investigation** (if warranted). The ORA Compliance Officer, NASA IRB Chair, or designee may conclude that additional information is needed before recommending a course of action. Further investigation may include:
   a. Consultation with Medical Monitors, Safety Officers, or other experts as appropriate.
   b. Review of research records including IRB correspondence, IRB applications and forms, recruitment materials, consent documents, and/or study data or records.
   c. Follow-up with study personnel, support staff, and research participants
   d. Meeting with the convened NASA IRB
4. Federally funded projects may require additional reporting (See SOP-810).

Review Outcomes

1. Unanticipated problem involving risks to subjects or others (UPIRSO):
   a. The problem or event was unexpected,
   b. The problem or event was related or possibly related to the research, or use of an investigational drug or device, and
   c. The problem or event suggests greater risks to participants or others than was previously known or recognized.

2. Event is NOT an Unanticipated Problem (UPIRSO).
3. IRB actions for events include but are not limited to:
    a. Acknowledgement without further recommendation;
    b. A request for more information from the researcher;
    c. Changes to the protocol and/or informed consent form;
    d. A requirement to inform enrolled participants or to reconsent enrolled participants;
    e. A change in frequency of continuing review;
    f. Requiring a corrective action plan;
    g. Further inquiry into other protocols of a similar nature;
    h. Suspension or termination of the project; and/or
    i. Directed post-approval verification or compliance audits
    j. Education

Communication

1. The IRB will notify the PI in writing indicating the review outcome - acknowledgment, requests for clarification or additional information, and/or finding within 15 calendar days.
2. If the PI has concerns regarding the IRB's decisions and/or recommendations, he/she/they may submit those concerns to the IRB in writing including a justification for changing the IRB decisions. The IRB reviews the request and makes a final determination, communicated in writing.
3. Please see additional reporting requirements at SOP-810.

References

OHRP Guidance (https://www.hhs.gov/ohrp/regulations-and-policy/guidance/reviewing-unanticipated-problems/index.html)

14CFR1230

21CFR50, 56

Version	Version notes	Originator	Approved by	Date
1		J. Ensley Gorshe/J. Kisenwether	M. Covington	21Oct2024